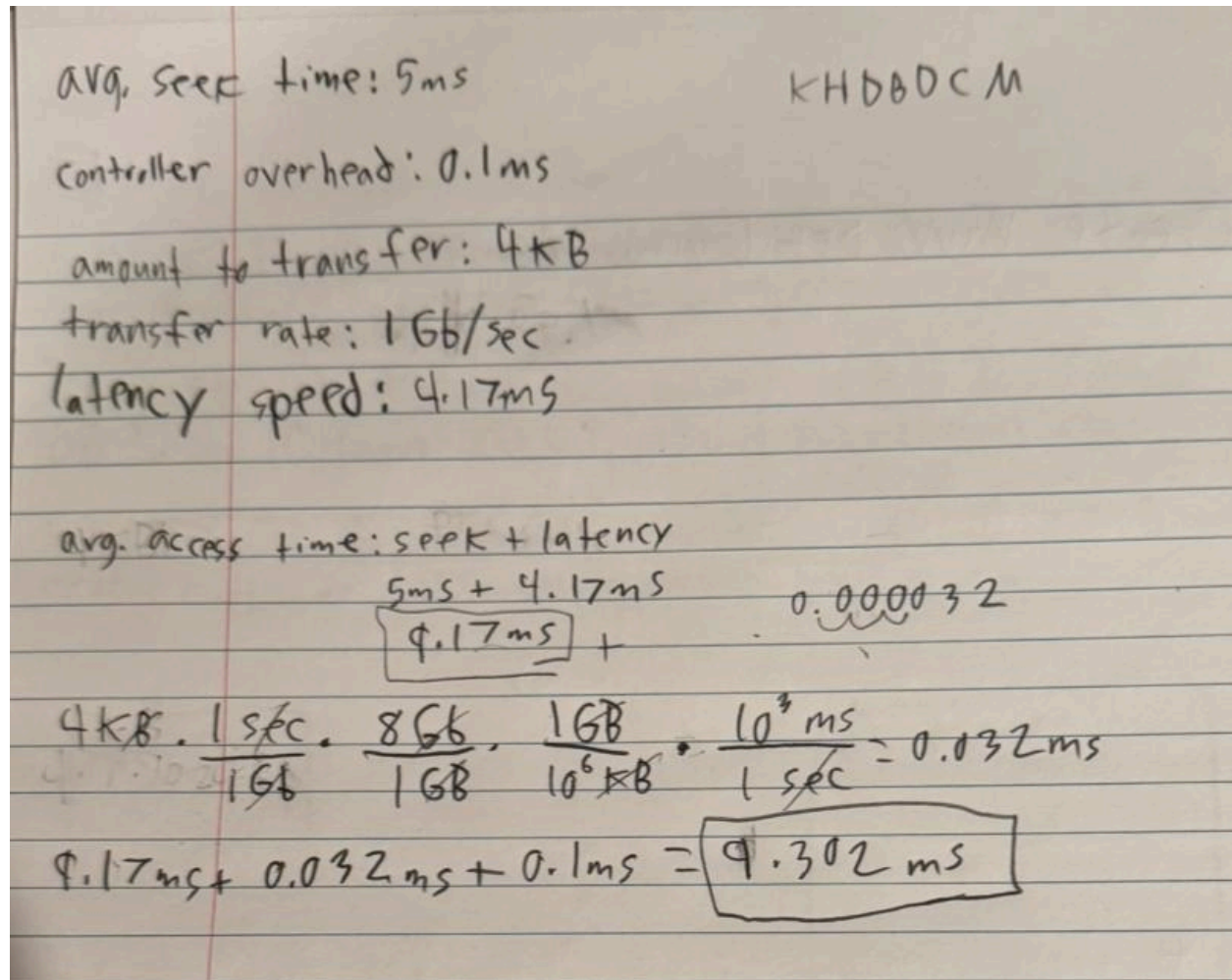


Calculate the time it takes (on average) to transfer a 4KB block on a 7200 RPM disk with a 5ms average seek time, 1Gb/sec transfer rate with a .1ms controller overhead (same example as in the book, please show your work):

9.302 ms



Handwritten calculation on lined paper:

avg. seek time: 5ms

controller overhead: 0.1ms

amount to transfer: 4KB

transfer rate: 1Gb/sec

latency speed: 4.17ms

avg. access time: seek + latency

$$5ms + 4.17ms = 9.17ms$$

0.000032

$$4KB \cdot \frac{1sec}{1Gb} \cdot \frac{8Gb}{1Gb} \cdot \frac{1Gb}{10^6KB} \cdot \frac{10^3ms}{1sec} = 0.032ms$$

$$9.17ms + 0.032ms + 0.1ms = 9.302ms$$

Calculate the time it takes (on average) to transfer a 8KB block on a 10,000 RPM disk with a 3ms average seek time, 1Gb/sec transfer rate with a .1ms controller overhead (please show your work):

6.164 ms

Amount to transfer: 8 KB

avg. latency: 3 ms

avg. seek time: 3 ms

transfer rate: 1 GB/sec

controller overhead: 0.1 ms

avg. access time = seek + latency

$$= 3 \text{ ms} + 3 \text{ ms}$$

$$= 6 \text{ ms}$$

$$\frac{8 \text{ KB}}{1} \cdot \frac{1 \text{ sec}}{1 \text{ GB}} \cdot \frac{8 \text{ GB}}{1 \text{ GB}} \times \frac{1 \text{ GB}}{10^6 \text{ KB}} \cdot \frac{10^3 \text{ ms}}{1 \text{ sec}} = 0.064 \text{ ms}$$

$$6 \text{ ms} + 0.064 \text{ ms} + 0.1 \text{ ms} = \boxed{6.164 \text{ ms}}$$

Calculate the time it takes (on average) to transfer a 64KB block on a 10,000 RPM disk with a 3ms average seek time, 1Gb/sec transfer rate with a .1ms controller overhead (please show your work):

6.612 ms

amount to transfer: 64KB

latency: 3ms

avg. seek time: 3ms

transfer rate: 1GB/sec

controller overhead: 0.1ms

$$\begin{aligned}\text{avg. access time} &= \text{seek time} + \text{latency} \\ &= 3\text{ms} + 3\text{ms} \\ &= 6\text{ms}\end{aligned}$$

$$\frac{64\text{KB}}{1} \cdot \frac{1\text{sec}}{1\text{GB}} \cdot \frac{8\text{GB}}{1\text{GB}} \cdot \frac{1\text{GB}}{10^6\text{KB}} \cdot \frac{10^3\text{ms}}{1\text{sec}} = 0.512$$

$$6 + 0.512 + 0.1 = 6.612\text{ms}$$